

Studio 5

INZBC AI OPERATING SYSTEM — BHANU GUPTA FINAL EVIDENCE PORTFOLIO

Final Evidence Portfolio

India New Zealand Business Council (INZBC) — Studio 5 Industry Placement

Studio 5 final evidence portfolio — Bhanu Gupta (100012452)

AUGUST 20, 2026

Table of Contents

1. Executive Summary	3
2. Introduction	3
2.1 Overview of the Project	3
2.2 Goals and Objectives	4
3. Project and Role Context	4
4. Learning Outcomes Covered	5
5. Project Requirements	5
5.1 Tools	5
5.2 Methodology	5
6. Evidence Against the Six PDR Objectives	6
7. Contributions to the Project / What Has Been Delivered (by Feature)	7
7.1 Foundation, Contracts and Architecture	7
7.2 Application Services and User-facing Contracts	7
7.3 Security and Trust Controls	8
7.4 Review and Integration Stewardship	8
7.5 Cross-repository Delivery	9
8. Evaluation of Problems and Technologies	11
8.1 A Worked Example of Objective 2.1	11
9. Communication for Sustainable Productivity	12
10. Managing the Project	13
10.1 Metrics Summary	13
10.2 Board Status and Reconciliation	13
11. Issues and Challenges / What's Left / Current Risks	14
11.1 Outside My Control	14
11.2 In My Control, In Progress	14
11.3 Learning Outcomes and Evidence to Sustain	14
12. Reflection	14
13. Conclusion	15
14. References	15
Appendix A — Evidence Portfolio (Screenshots)	16
A.1 — Team and Client Communication (Objective 3.1)	16
A.2 — Evidence Register	16

1. Executive Summary

I served as technical lead for foundation, security, review, and cross-repository integration on the INZBC AI Operating System and Digital Platform. My work converted the Council's core safeguards—sourced trade evidence, named human approval, privacy, separation of duties, and accountable decisions—into shared contracts, database constraints, API controls, user-interface states, automated tests, CI gates, and operational documentation.

The refreshed GitHub evidence records 312 attributable default-branch commits across four organisation repositories, 135 authored pull requests (134 merged), 141 authored issues, 60 review events across 45 pull requests, and 187 of the organisation's 189 merges performed by me. These figures are paired throughout this report with representative controls, review outcomes, screenshots, and live board state so volume is never used as a substitute for quality.

Overall assessment: all six Studio 5 PDR objectives are strong and fully demonstrated. Bhanu's technical delivery, analytical judgement, team communication, collaboration, project management, documentation, technical leadership, and integration work were all strong and are supported by attributable evidence.

Evidence area	Verified result	Why it is strong
Delivery	312 commits; 134 merged authored PRs	Sustained, cross-repository, production-oriented contribution
Review and integration	60 reviews; 187 merges	Consistent human quality gate and correction loop
Planning and traceability	141 issues; 79 board cards	Requirements, dependencies, blockers, and acceptance criteria remained visible
Communication and teamwork	Team/client calls, issues, reviews, ADRs, worklogs	Cross-lane decisions and client dependencies were communicated in durable forms

2. Introduction

2.1 Overview of the Project

The project is a controlled digital platform for the India New Zealand Business Council. It combines a Strategic Intelligence Platform (SIP) for reviewed trade briefs, an FTA information and explainer service, an AI-assisted communications workflow, and a refreshed public website. The three-student team comprised Bhanu Gupta, Roshan Aryal, and Paras, working with INZBC and course supervisors during Studio 5.

The core challenge was organisational trust, not simply feature delivery. Trade statements had to point to stable evidence or return an explicit no-match result; AI-assisted content had to remain draft-only until a named human approved it; sensitive data required minimisation and redaction; and operational decisions needed an audit trail that could not be rewritten after the fact.

2.2 Goals and Objectives

- Give INZBC a maintainable technical foundation with documented boundaries, reproducible environments, shared contracts, and a credible handover path.
- Keep every AI-assisted workflow under named human review and make prohibited or unauthorised actions fail closed.
- Return sourced, dated trade and FTA answers, including an explicit no-match outcome when the evidence does not support a claim.
- Protect member and operational data through server-side roles, CSRF protection, separation of duties, redaction controls, and append-only evidence records.
- Use issues, pull requests, review gates, CI, decision records, and the project board to create an auditable delivery history.
- Leave documentation that enables INZBC or a future team to understand, operate, test, and extend the platform.

3. Project and Role Context

The team divided ownership by workstream while keeping contracts and reviews shared. My role was the technical and integration boundary: I owned or co-owned the database schema, API contracts, authentication and control-plane services, model gateway, SIP core/orchestrator, security invariants, CI quality gates, operational documentation, and merge stewardship. Roshan built intelligence, sources, data, FTA, collection, and service-side features against those contracts. Paras owned product and application-layer delivery, including the public website, member-facing pages, the Communications Assistant interface, and the SIP review UI.

Responsibility	Bhanu's accountability	Team interface
Foundation and contracts	Architecture, schema, OpenAPI, generated types, environment and repository rules	Published boundaries used by Roshan and Paras
Security and control plane	Auth/RBAC, CSRF, redaction/model boundary, separation of duties, append-only records	Protects every service and user-facing workflow
Review and integration	CI gates, code review, correction requests, controlled merges, cross-repo reconciliation	Maintains one quality standard across team lanes
Delivery governance	Issues, ProjectV2 fields, ADRs, worklogs, operator and handover documents	Connects technical state to team and client decisions

4. Learning Outcomes Covered

The portfolio covers Learning Outcomes 1–3 through the six PDR objectives used in the Studio 5 assessment.

LO	Learning outcome	Mapped objectives	Bhanu judgement
1	Analyse real-world problems and create IT systems to solve them	1.1, 2.1	Strong — fully demonstrated
2	Apply technical and theoretical skills to unfamiliar and complex situations	1.2, 2.1	Strong — fully demonstrated
3	Contribute meaningfully as a team member in a structured development process	1.2, 3.1, 3.2, 3.3	Strong — fully demonstrated

5. Project Requirements

5.1 Tools

- Backend and contracts: Python 3.11+, FastAPI, Pydantic, PostgreSQL, SQL, OpenAPI, and generated TypeScript client types.
- Frontend and web: React, TypeScript, Vite, Tailwind CSS, Astro, HTML/CSS/JavaScript previews, and Wix integration material.
- Version control and planning: GitHub repositories, protected pull-request workflow, CODEOWNERS, issue templates, GitHub Actions, and ProjectV2.
- Quality and security: pytest with coverage, Ruff, frontend lint/typecheck/test/build, Semgrep, gitleaks, actionlint, dependency review, link checks, Docker builds, and endpoint smoke tests.
- Communication and documentation: issues, pull requests, ADRs, workstream logs, meeting records, WhatsApp/client coordination, Google Drive references, and Moodle reporting.

5.2 Methodology

I used a contract-first, issue-led, iterative development method. Each task began with context, scope, acceptance criteria, dependencies, lane, priority, and phase. Work then moved through a focused branch and pull request, automated verification, human review, correction where required, and controlled merge. Cross-lane changes were checked against the owning workstream and the published API or schema rather than relying on verbal assumptions.

Security-sensitive requirements were implemented as server-side and database invariants and tested at their real boundary. Client-owned choices—identity, membership integration, named reviewer, organisation secrets, OAuth ownership, and production responsibility—remained explicit board items rather than being guessed by the placement team.

6. Evidence Against the Six PDR Objectives

#	Objective	Assessment	Evidence and judgement
1.1	Capture requirements using an appropriate methodology and tool/platform.	Strong — fully demonstrated	Contract-first requirements, ADRs, project charter, API/schema specifications, issue templates, acceptance criteria, and ProjectV2 fields created a durable trace from need to merge.
1.2	Contribute meaningfully to the project/codebase.	Strong — fully demonstrated	312 default-branch commits and 134 merged authored PRs across four repositories delivered architecture, contracts, services, security controls, documentation, public-site concepts, and operational tooling.
2.1	Evaluate problems and technologies, research independently, and justify decisions.	Strong — fully demonstrated	FastAPI/OpenAPI, PostgreSQL constraints, fail-closed model boundaries, generated clients, CI/security gates, and multi-repo boundaries were evaluated against enforceability, testability, privacy, and handover needs. The worked example in section 8.1 shows the reasoning chain.
3.1	Communicate within a team to maintain sustainable productivity.	Strong — fully demonstrated	141 scoped issues, 60 review events, weekly team/client calls, written decision packs, ADRs, worklogs, meeting records, and cross-lane contract support show frequent, actionable, and durable communication.
3.2	Use industry-standard communication and project-management tools professionally.	Strong — fully demonstrated	GitHub issues/PRs/Actions/Projects were used throughout. Bhanu held 79 assigned board cards, reconciled stale status fields against repository state, and used a safe board-editing path.
3.3	Document all project aspects, including user and technical documentation.	Strong — fully demonstrated	Architecture/data-model diagrams, API and integration specifications, operator guide, security/privacy registers, incident response, backup/monitoring, migration/rollback, governance/RACI, worklogs, and client-decision records support development and handover.

PDR result: strong / fully demonstrated in 1.1, 1.2, 2.1, 3.1, 3.2, and 3.3. In particular, team communication and sustainable productivity are strongly evidenced through both live meeting screenshots and a substantial written review, issue, decision, and handover trail.

7. Contributions to the Project / What Has Been Delivered (by Feature)

My contribution spans platform foundation, security, application services, delivery governance, code review, and public-site experimentation. The headline metrics below are current organisation evidence, while the subsections identify the deliverables represented by that activity.

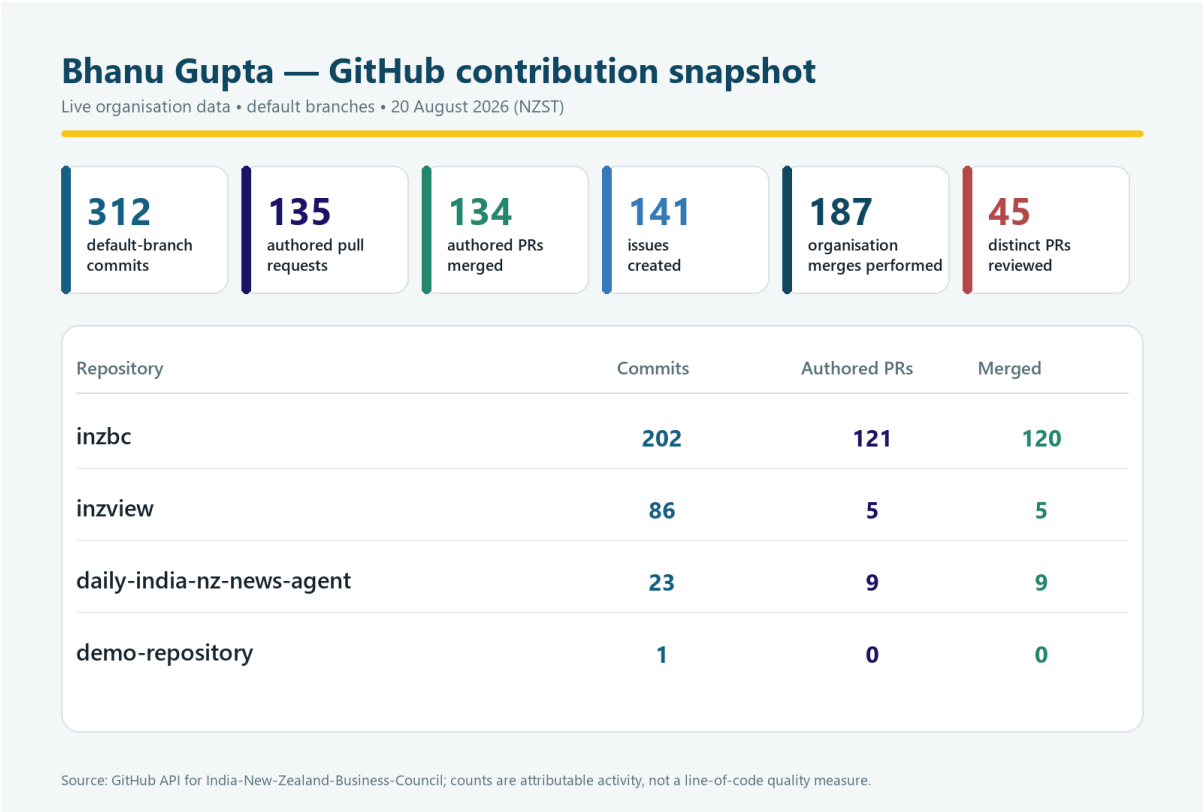


Figure 1. API-derived GitHub contribution snapshot for Bhanu Gupta across all four organisation repositories.

7.1 Foundation, Contracts and Architecture

I established the initial AI architecture and repository operating model, added the README, contribution rules and pull-request templates, scaffolded the monorepo, and published shared contracts. I then added the SIP-185 mandatory source register keyed by stable identifiers, the model gateway and SIP-050 scoring path, the fail-closed SIP run state machine, system architecture and data-model diagrams, and an operator guide. Representative evidence includes PRs #1, #3, #16, #27, #34, #67, #71, and #72.

7.2 Application Services and User-facing Contracts

I helped move the project from documents and prototypes into an integrated FastAPI/PostgreSQL service and generated frontend contract. Work included the distinct FTA no-match outcome, the FTA query endpoint, React query/escalation states, public API hardening, report submit/read endpoints, dashboard data, audit reads, persisted orchestrator rehydration, and GitHub sign-in. Representative PRs are #83, #85–#87, #240, #296, #302, #305, #308, and #311.

7.3 Security and Trust Controls

The implementation refuses model use until an approved redaction policy exists, rejects prohibited data even if a masking rule could hide it, enforces server-side roles, checks separation of duties against recorded acts, requires CSRF tokens for writes, and records decision and audit evidence as append-only events. Semgrep and gitleaks are blocking CI gates. Representative PRs are #164, #180, #224, #228, #272, #278–#279, #284–#285, #300, #306–#307, and #315–#316.

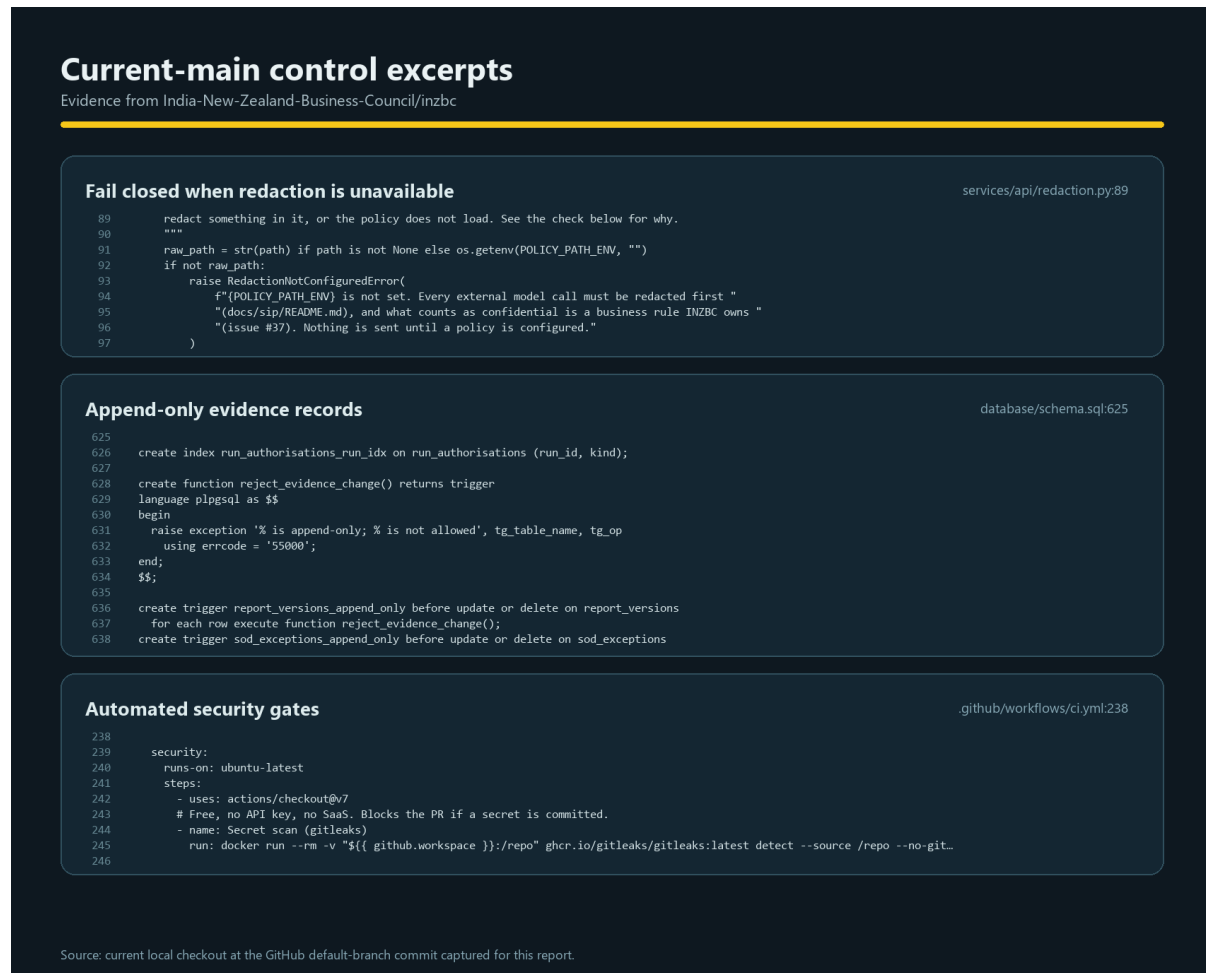


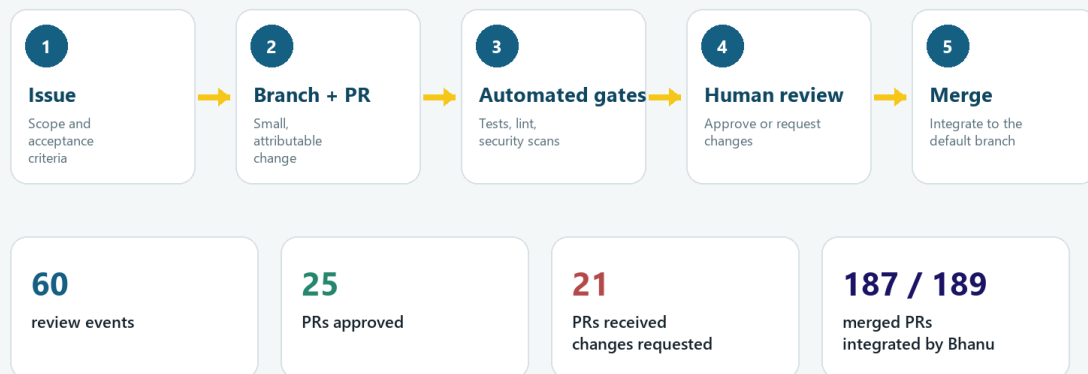
Figure 2. Current-main excerpts showing fail-closed redaction, append-only records, and automated security gates.

7.4 Review and Integration Stewardship

I used the shared pull-request queue to challenge correctness, security, tests, contracts, documentation, and operational claims before integration. The refreshed history contains 60 review events across 45 pull requests: 25 received approval and 21 received at least one changes-requested review. These sets overlap because corrected work was reviewed again. I performed 187 of 189 organisation merges, providing a consistent integration gate across team lanes.

Quality-gate and integration stewardship

Repository evidence for review, revision and controlled integration



Approval and changes-requested sets overlap when a pull request went through more than one review round.

Source: GitHub GraphQL API review and merge records, captured 20 August 2026.

Figure 3. Issue-to-merge quality gate and Bhanu's review/integration evidence.

7.5 Cross-repository Delivery

In *inzview* I authored five merged pull requests and contributed 86 default-branch commits covering cinematic and future-facing homepage concepts, visual effects reels, and motion/readability fixes. In *daily-india-nz-news-agent* I authored nine merged pull requests and contributed 23 default-branch commits covering governance, controlled launch/distribution, a branded brief, scheduling and backup changes, CI/security checks, and controlled links to SIP documentation. The demo repository contains only my initial commit and is included solely for organisation-wide completeness.

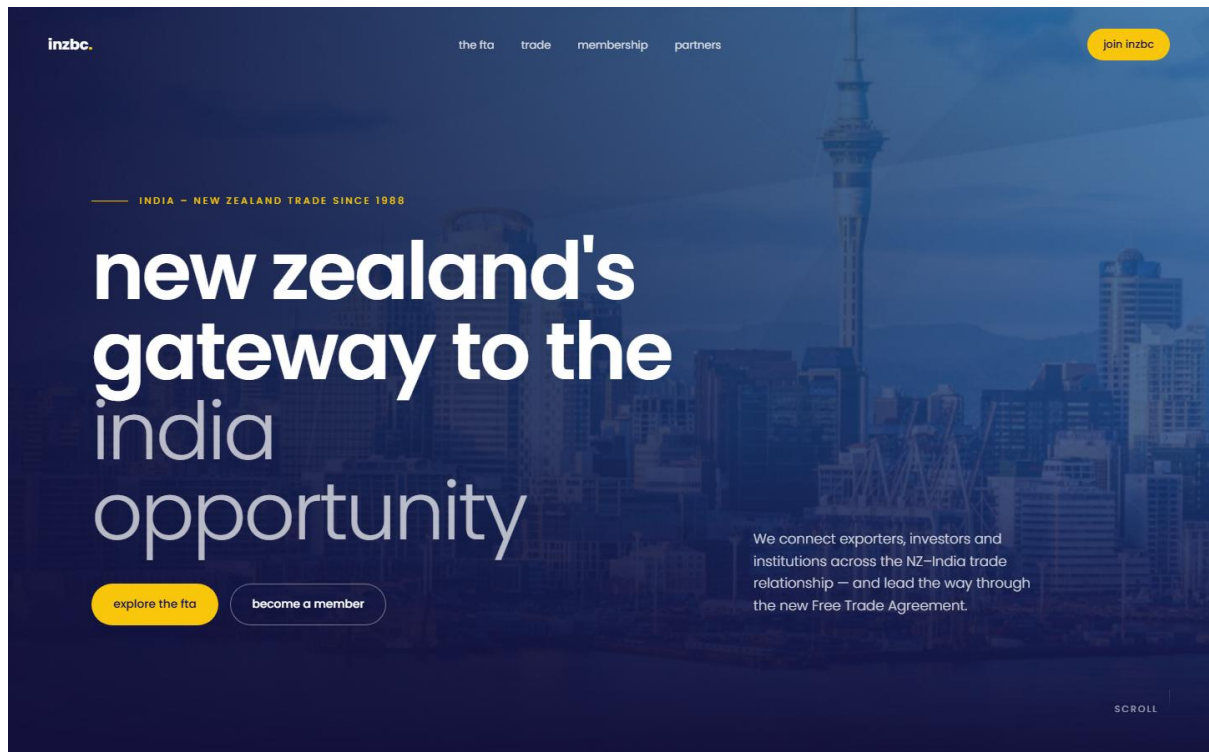


Figure 4. Cinematic inzbz homepage concept rendered from the refreshed private main branch.

8. Evaluation of Problems and Technologies

The central problem was not merely generating content. INZBC needed a system that could use trade evidence without inventing facts, protect member and operational data, preserve accountable human decisions, and remain supportable after placement. I evaluated technology against four properties: explicit contracts, enforceable safety rules, testability at the real boundary, and credible operational handover.

FastAPI, Pydantic, and OpenAPI. These technologies let runtime validation, published API documentation, generated TypeScript types, and contract-drift checks share one source. The trade-off is a deliberate regeneration obligation whenever the schema changes; CI enforces it.

PostgreSQL. Transactions, foreign keys, row locking, append-only triggers, CHECK constraints, and role grants express audit, concurrency, and separation-of-duties rules below the UI. This adds migration, backup, and recovery responsibility, which is why operational documents were treated as product work.

React, TypeScript, Vite, Astro, and Wix. React and generated types suited interactive SIP/FTA/comms workflows; Astro and isolated HTML previews suited public-site composition and motion experiments. Separate repositories protected the controlled SIP service from website experimentation, at the cost of additional coordination and shared-brand work.

GitHub Actions and ProjectV2. CI catches repeatable defects and Projects makes scope/blockers visible, but neither replaces human judgement about business rules, evidence quality, privacy, or client authority. The review history and board reconciliation show both the value and limits of these tools.

8.1 A Worked Example of Objective 2.1

Case: enforcing safe AI and approval boundaries. A prompt, disclaimer, or disabled button would be easy to bypass and difficult to audit. Redaction alone was also insufficient because masking prohibited information could still permit an unauthorised model call. I therefore evaluated where the control had to live and chose layered, fail-closed enforcement.

- Single boundary: all model calls pass through one server-side gateway, allowing policy and usage checks to be applied consistently.
- Fail closed: the gateway refuses a call when no approved redaction policy exists and rejects prohibited data rather than trying to hide it.
- Authorised writes: session roles and CSRF tokens protect business mutations; gated actions also check separation of duties against recorded acts.
- Immutable evidence: decision and audit records are append-only in PostgreSQL so the history cannot be rewritten through an application bug or ordinary role.
- Independent verification: unit/integration tests plus Semgrep and gitleaks gates exercise the actual boundary. PRs #34, #164, #272, #284, #300, #315, and #316 show the decision's evolution.

The result is a principled system decision rather than a presentation-layer promise: the same business safeguard is expressed in the API, database, UI state, tests, review record, and operator documentation. That is why Objective 2.1 is assessed as strong and fully demonstrated.

9. Communication for Sustainable Productivity

Communication was a strong part of my contribution. I translated broad requirements into 141 scoped issues with context, dependencies, and acceptance criteria; used 60 review events to approve sound work or request concrete corrections; and supported cross-lane delivery through published contracts, ADRs, workstream logs, meeting records, operator material, and client-decision packs. This created a durable record alongside day-to-day calls and WhatsApp coordination.

The review trail shows collaborative quality control rather than passive merging. Changes-requested reviews identified security gaps, contract drift, unsupported claims, missing tests, and handover problems before integration. Because corrected pull requests could then be approved, the record demonstrates responsiveness and a sustainable feedback loop.

Client-dependent questions were framed with consequences: identity affects every protected page and the separation-of-duties model; a named reviewer determines whether AI-assisted content can progress; Member Jungle determines the membership boundary; and organisation-owned secrets affect production. The screenshots in Appendix A show Bhanu participating in team and client calls, while the repository trail shows how those discussions were converted into actionable written work.

Communication judgement: strong — frequent team and client engagement, clear technical translation, responsive code review, documented decisions, and handover-oriented writing all supported sustainable team productivity.

10. Managing the Project

10.1 Metrics Summary

Metric	Current value	Interpretation
Default-branch commits	312	Attributable across all four organisation repositories
Authored pull requests	135 total / 134 merged	Focused delivery with one current open dependency update
Issues authored	141	Requirements, work breakdown, dependencies, and client blockers
Human review	60 events / 45 PRs	Approvals and correction requests across team work
Merge stewardship	187 of 189 merges	Consistent integration responsibility
Assigned board footprint	79 of 138 cards	Platform, product, process, client, and intelligence lanes

10.2 Board Status and Reconciliation

The private INZBC SIP Platform board contains 138 items. My 79 assigned cards display 32 Done, 2 In Progress, 9 Blocked, and 36 Todo. They comprise 54 issue cards and 25 pull-request cards.

I reconciled those fields against the underlying repository state. All 25 assigned pull-request cards were merged; 32 issue cards were closed and 22 remained open. However, 25 closed issues still displayed Todo or Blocked. This status drift understates delivery and weakens forecasting, so the report preserves both the displayed field and the underlying state rather than silently rewriting history.

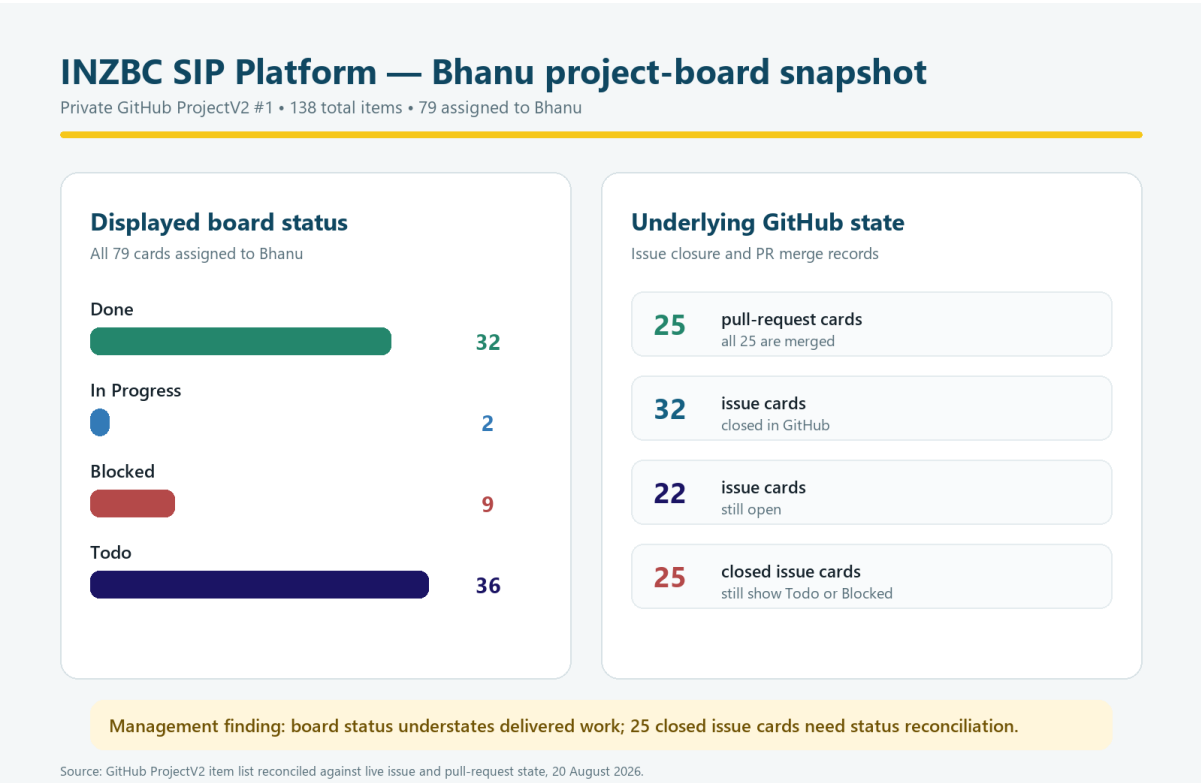


Figure 5. Bhanu's ProjectV2 status snapshot reconciled against live issue and pull-request state.

11. Issues and Challenges / What's Left / Current Risks

11.1 Outside My Control

- INZBC must choose the identity/login model, production OAuth ownership, and long-term operating owner before protected production access can be claimed.
- INZBC must confirm the membership-system boundary, including whether and how Member Jungle is retained or integrated.
- A named human reviewer and final approval responsibilities must be confirmed before AI-assisted outputs can move beyond controlled drafts.
- Organisation-owned secrets and rotation policy are required for production integrations and live end-to-end runs.

11.2 In My Control, In Progress

- Complete and evidence the SIP-050 evaluation harness and the SIP-184 orchestrator against golden-set, regression, prompt-injection, and hard human-gate cases.
- Keep dependency and quality gates current, including the tracked Ruff upgrade, without weakening locked and repeatable builds.
- Run routine ProjectV2 reconciliation so closed/merged work and displayed status do not diverge.
- Continue review and handover support while distributing integration knowledge so the quality gate is sustainable beyond one person.

11.3 Learning Outcomes and Evidence to Sustain

All three learning outcomes and all six objectives are already strongly evidenced. The remaining responsibility is to sustain the same cadence through handover: keep CI green, preserve the written decision trail, close or explicitly carry forward client dependencies, reconcile the board, and demonstrate the final operator path with a named human reviewer.

12. Reflection

The most important learning was that security and governance are strongest when they are business invariants, not review reminders. A statement such as “nothing publishes without a named reviewer” is incomplete until the API, role model, database, UI, test suite, and operator guide all make the prohibited path impossible or visible.

I also learned to distinguish repository state from reporting state. The board's 25 closed-but-not-Done issue cards show that good engineering can be obscured by stale management fields. Reconciliation must therefore be part of the definition of done, and evidence reports should preserve discrepancies rather than selecting whichever number looks better.

My review and merge role was a strength because it gave the team one integration standard, but it also creates concentration risk. The next improvement is to make review criteria, release checks, and operational knowledge easy for another maintainer to apply. Strong leadership is not keeping all control; it is leaving controls that remain effective when someone else owns them.

13. Conclusion

My contribution is best described as strong technical foundation and integration leadership. I converted client safeguards into enforceable contracts and controls, helped deliver working services and public-site concepts across four repositories, maintained a rigorous review and merge gate, translated meetings and uncertainties into actionable written work, and produced documentation for continued operation.

The evidence supports a strong judgement in every assessed area: 312 attributable default-branch commits, 134 merged authored pull requests, 141 issues, 60 review events, 187 organisation merges, live team/client coordination, and a broad technical and handover record. Technical delivery, teamwork, communication, project management, documentation, analytical reasoning, and leadership were all strong from Bhanu Gupta.

The remaining work is explicit rather than hidden: complete the two active Must platform items, resolve client-owned identity/reviewer/membership/secrets decisions, reconcile board fields, keep the quality gates green, and complete operational handover. Those actions move a strong controlled implementation toward an authorised and sustainable INZBC service.

14. References

- [India New Zealand Business Council GitHub organisation](#)
- [INZBC AI Operating System repository](#)
- [inzview public-site repository](#)
- [Daily India–NZ news agent repository](#)
- [INZBC SIP Platform ProjectV2 #1](#)
- [Bhanu workstream record](#)
- [INZBC role and product overview](#)
- [Project charter and non-negotiable rules](#)
- [PR #300 — refuse prohibited data at the model boundary](#)
- [PR #272 — session auth, roles, and separation of duties](#)
- [PR #315 — CSRF token on SIP UI writes](#)
- [FastAPI documentation](#)
- [PostgreSQL documentation](#)
- [GitHub Actions documentation](#)
- [GitHub Projects documentation](#)

Evidence note: GitHub REST, GraphQL, search, and private ProjectV2 data were refreshed on 20 August 2026 through the authenticated Bhanu Gupta account. Roshan's and Paras's supplied reports were used for section coverage and cross-team evidence; any screenshot reused from them is identified in its caption.

Appendix A — Evidence Portfolio (Screenshots)

A.1 — Team and Client Communication (Objective 3.1)

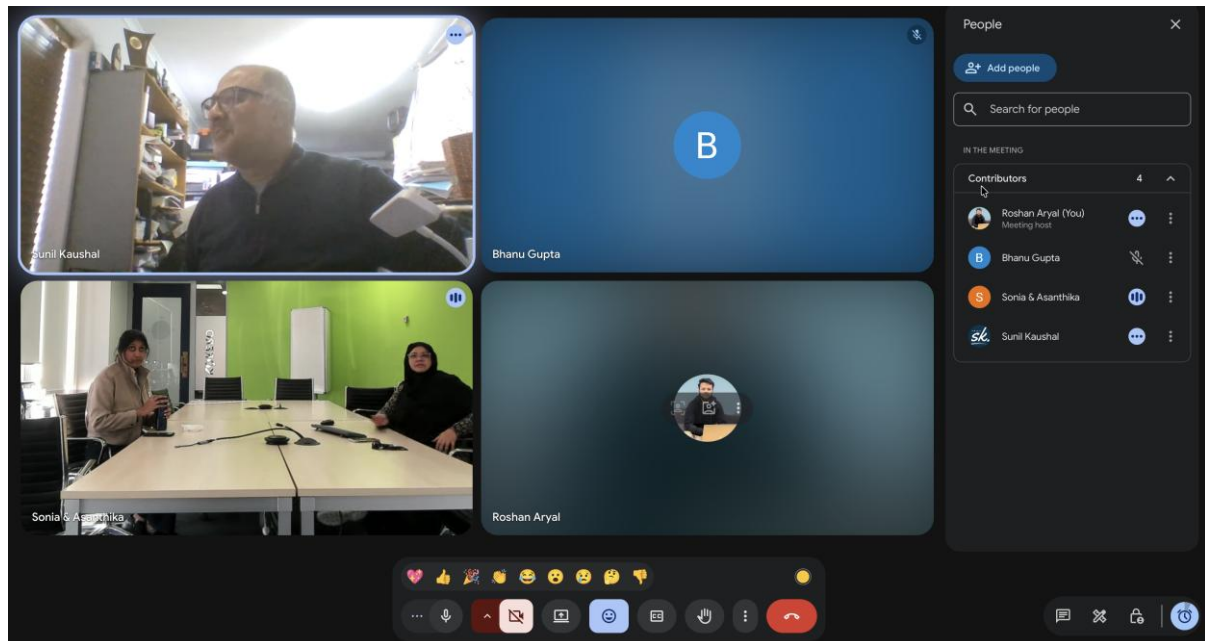


Figure A1. Client coordination meeting supplied in Roshan's evidence portfolio: Sunil Kaushal, Bhanu Gupta, Sonia and Asanthika, and Roshan Aryal.

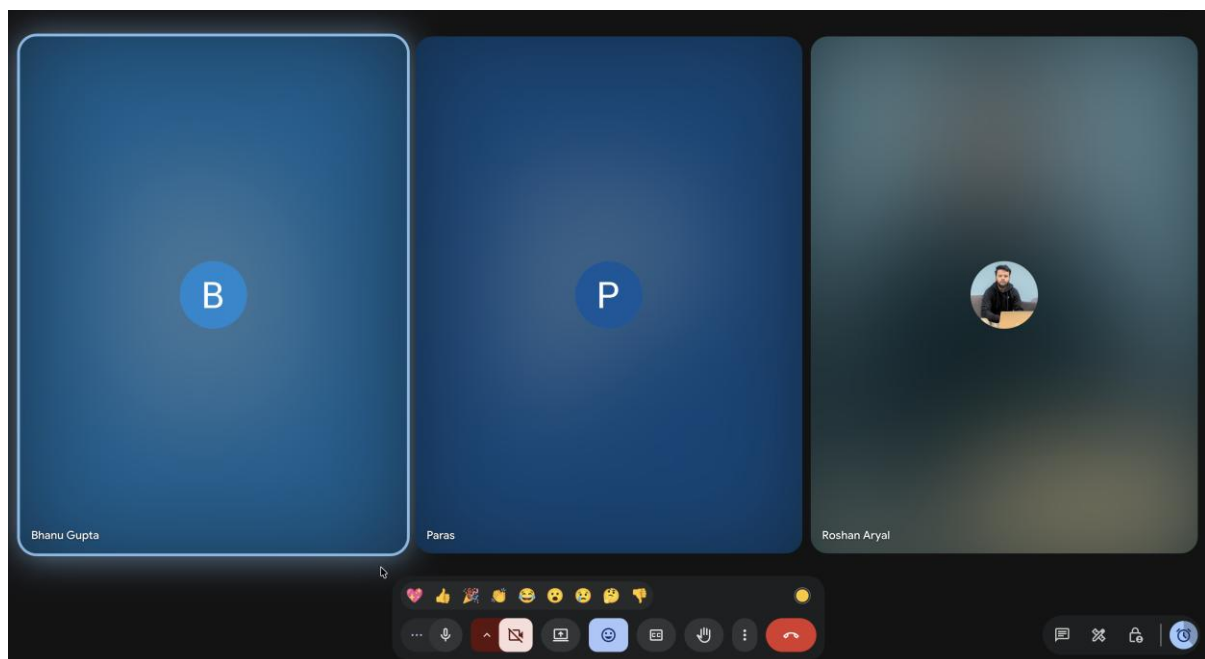


Figure A2. Team coordination call supplied in Roshan's evidence portfolio: Bhanu Gupta, Paras, and Roshan Aryal.

A.2 — Evidence Register

Evidence	Supports	Source/scope
Figure 1 — contribution snapshot	1.2, 3.2	Refreshed GitHub organisation data
Figure 2 — current-main controls	1.1, 2.1, 3.3	Current inzbc main source excerpts

Evidence	Supports	Source/scope
Figure 3 — quality gate	1.2, 3.1, 3.2	Complete PR review/merge history
Figure 4 — inzview render	1.2, 2.1	Private inzview main browser rendering
Figure 5 — board reconciliation	3.2, 3.3	Private ProjectV2 and repository state
Figures A1–A2 — meetings	3.1	Roshan-supplied team/client screenshots